IE540
DP (Dynamic Programming) & RL (Reinforcement Learning)

2025 Spring

Hayong Shin
KAIST, Dept. of ISE

Course logistics

- Class hour
  - Tue/Thu 10:30~11:45am
    - classroom : E2-2, #B105 (Basement floor)
  - lecture videos will be available on KLMS and KOOC

- Instructor : Prof. Hayong Shin
  - Tel : 042-350-3124
  - Email : hyshin@kaist.ac.kr
- T.A.
  - Merve Doganbas (Tel : 042-350-3164, mervedoganbas@kaist.ac.kr)
- Course home page
  - http://klms.kaist.ac.kr
    - Syllabus, honor code pledge, survey
    - Lecture notes, Lecture video, supplementary materials incl. python code (*.ipynb)
    - Quizzes, Homework submission, Grading results
    - Bulletin board : Q & A, Wishes, Discussion, Other issues
- Text book
  - Sutton & Barto, Reinforcement Learning : An Introduction, 2nd ed., MIT Press, 2018
    - http://incompleteideas.net/book/the-book.html
  - D.Bertsekas, Reinforcement Learning and Optimal Control, Athena Scientific, 2019

Grading policy (subject to change during the course)

- Class participation [30%]
  - Quizzes : 15%
  - Q & A, Discussion : 15%
    - KLMS Q&A board activity
    - in class Q&A, interaction/response, discussion
- Exam [30%] : final exam only
  - exam type will be announced later
- Assignments & projects [40%]
  - Programming assignments [20%]
    - Violating academic integrity (copying homework)
      - 2 letter grade will be lowered for the first time detected (A+ → A-)
      - Detected 2nd time will make it a failure
    - Submit using the course homepage at KLMS
      - 20% of delay penalty / day
  - Term project [20%]
    - 2~1 students / team
      - Duo is recommended, solo is fine.
        - » Trio will be exceptionally allowed if the topic is big.
      - default project : cartpole competition
      - your own project : welcome

[Tentative] Course Schedule

Date (Tue)	contents	Date (Thu)	contents
2/25	Course Introduction, L1.Intro. to RL	2/27	L2.MDP
3/4	L3. DP methods	3/6	Quiz, Q&A on L1, L2, L3 HW #1 : DP for Gridworld
3/11	L4. Monte-Carlo methods	3/13	Quiz, Q&A on L4
3/18	L5. TD methods	3/20	Quiz, Q&A on L5 HW #2 : MC/TD control for Gridworld
3/25	L6. n-Step TD	3/27	Quiz, Q&A on L6
4/1	L7. Value function approximation	4/3	Quiz, Q&A on L7. VFA
4/8	L8. DQN, L8-1. NN	4/10	Quiz, Q&A on L8. DQN HW #3: DQN for Cartpole
4/15	mid-term exam week	4/17	mid-term exam week

[Tentative] Course Schedule

Date (Tue)	contents	Date (Thu)	contents
4/22	L9. Policy Opt. (SPG)	4/24	Quiz, Q&A on L9.
4/29	L10. Policy Opt. (TRPO, PPG)	5/1	Quiz, Q&A on L10. HW #4: SPG for Cartpole
5/6	No class (Video) L11. Policy Opt. (DPG, DDPG, CEM)	5/8	Quiz, Q&A on L11 HW #5: DDPG & CEP for Cartpole
5/13	L12. Multi-Armed Bandit	5/15	Quiz, Q&A on L12 (MAB)
5/20	L13. Average reward, Finite horizon	5/22	Quiz, Q&A on L13
5/27	L14. Reward shaping, AlphaGo	5/29	Quiz, Q&A on L14
6/3	Term project result submission	6/5	(Online) Term project poster exhibition
6/10	Final exam week	6/12	Final exam week

Course format : Video + Real-time (in-person or zoom) mixed

- Video lecture (video + slide)
  - video will be open anytime after upload
    - you must watch video before the next real-time class
      - videos will be available on KLMS or KOOC (kooc.kaist.ac.kr/reinforcement)
    - there will be a quiz at the beginning of the following real-time class
  - pre-recorded lecture videos are very condensed
    - pause → think → resume
      - fill in blanks (DIY boxes), think ahead questions (TAQ boxes)
    - video lectures are mostly given in Korean with English subtitles
    - post your own questions on KLMS board
      - class participation credit!!
- Real-time class (mostly on Thursday class hour)
  - in-person or online(zoom) class
  - quiz for 5 min
    - via KLMS - bring a laptop, a tablet, or a smartphone
    - 2 allowances (worst 2 quizzes will be removed from scoring)
  - answer to questions : TAQ's & questions posted by students
  - interactive demo
  - homework assignment

Some useful links

- RL
  - RLAI incompleteideas.net/rlai.cs.ualberta.ca/RLAI/rlai.html
  - Algorithms for Reinforcement Learning, Szepesvari
    - www.ualberta.ca/~szepesva/papers/RLAlgsInMDPs.pdf
  - GridWorld demos
    - REINFORCEjs cs.stanford.edu/people/karpathy/reinforcejs/gridworld_dp.html

- Python documents
  - docs.python.org/3/index.html (tutorial & reference are in KLMS)
  - Matplotlib.org (Plotting tool)
  - riverbankcomputing.com/software/pyqt/intro (PyQt)

- PyTorch : www.pytorch.org
- TensorFlow : www.tensorflow.org
- OpenAI : www.openai.org

Preparing programming environment

- Installing python : Anaconda or Miniconda
- IDE for python : PyCharm or jupyter lab
- MatPlotLib : plotting library for python
- PyQt : Building GUI
- OpenAI Gym : collection of RL environments
  - to be introduced later

Installing Python (Anaconda or Miniconda)

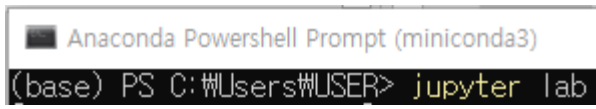
- Install required softwares
  - Anaconda : https://www.anaconda.com/products/individual
    - Comprehensive python development set including:
      - Python interpreter (multiple versions such as 2.7, 3.8),
      - Other popular modules : numpy, matplotlib, Pyqt, IDLE, Spyder, Jupyter, …
  - Miniconda : https://docs.conda.io/en/latest/miniconda.html
    - Minimum set : numpy, PyQt, …
      - Fast
    - Need to install required packages
      - conda install package_name
      - scipy
      - scikit-learn
      - Matplotlib
      - jupyter
      - PyTorch or TensorFlow

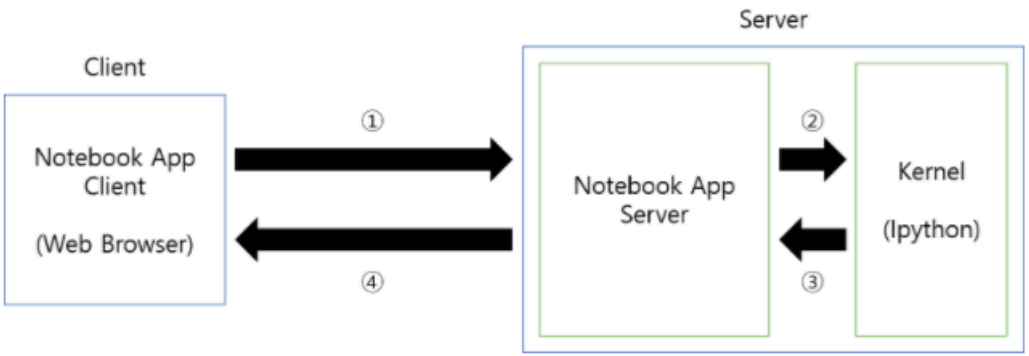
Windows installers

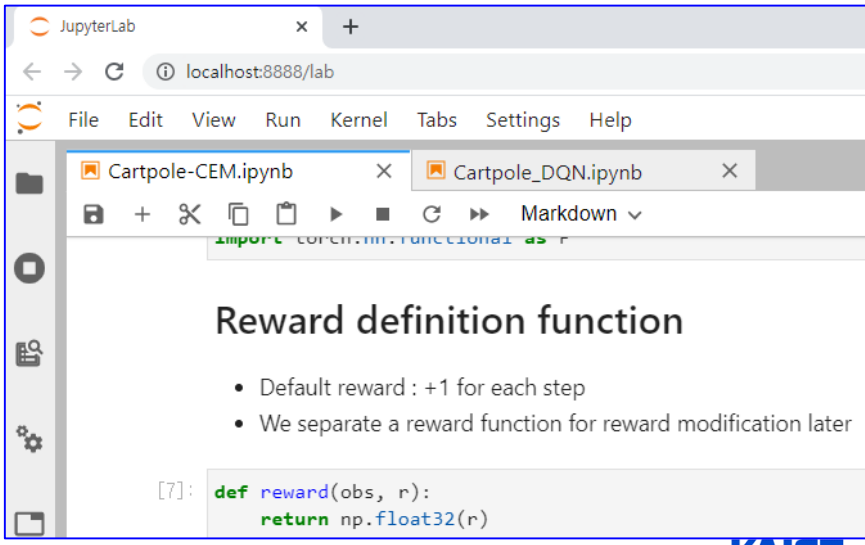
Python version	Name
Python 3.8	Miniconda3 Windows 64-bit
	Miniconda3 Windows 32-bit
Python 2.7	Miniconda2 Windows 64-bit
	Miniconda2 Windows 32-bit

IDE for python

- IDE (Integrated Development Environment)
  - Editor, Debugger, …
  - PyCharm : http://www.jetbrains.com/pycharm/download/
    - Community edition for free
  - IDLE, Spyder, Visual Studio, Eclipse
- Jupyter Notebook
  - Not quite an IDE, but an interactive environment
    - Very useful and popular, Run cell by cell, Markdown, …
  - install jupyter lab : pip install jupyterlab
  - run jupyter lab server : "jupyter lab" in Anaconda Command Prompt
    - jupyter lab on web browser
  - Debugging?



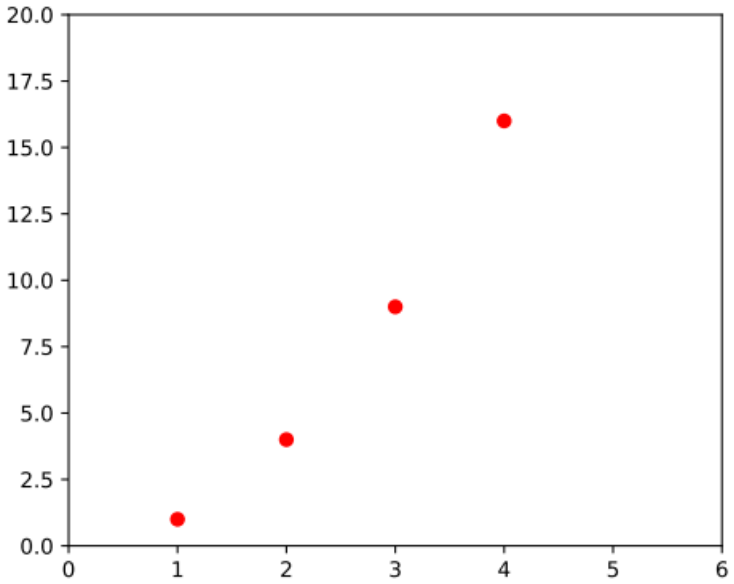




Matplotlib

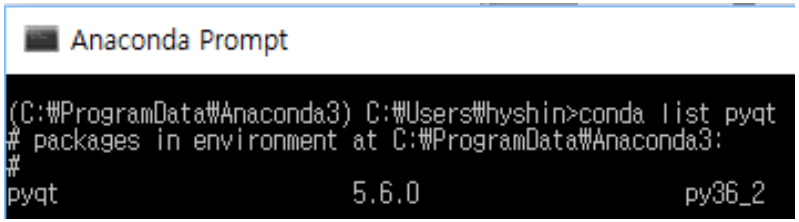
- Matplotlib.org
  - Plotting library for python
  - Similar to Matlab plot
  - Install
    - conda install matplotlib
    - pip install matplotlib
  - Manual, example, …
    - http://matplotlib.org/2.0.0/contents.html

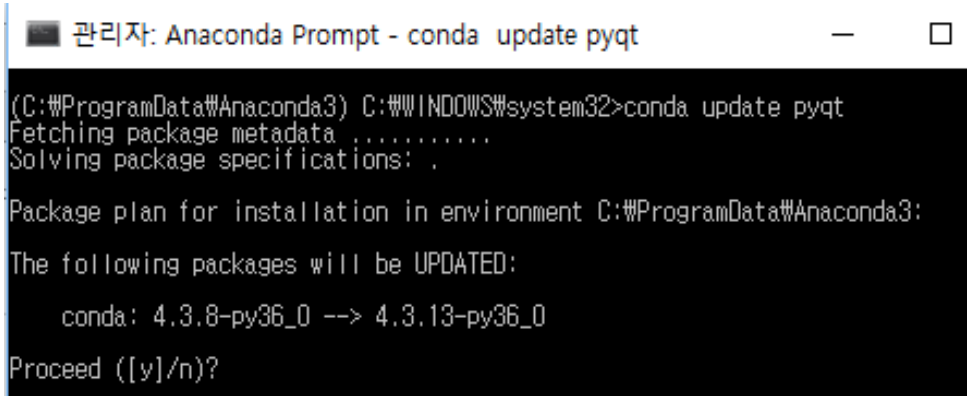
```
import matplotlib.pyplot as plt
plt.plot([1,2,3,4], [1,4,9,16], 'ro')
plt.axis([0, 6, 0, 20])
plt.show()
```



GUI using PyQt

- Preparation
  - Install Anaconda or Python interpreter
  - Install PyCharm
- Installing PyQt
  - www.riverbankcomputing.com/software/pyqt/download5
  - From PyCharm terminal or Anaconda console ("run as administrator")
    - pip install PyQt5
    - PyQt is already included in Anaconda
      - From Anaconda console conda list pyqt (check version)
        - » conda update pyqt (update to the latest version)
  - PtQt Reference Guide : http://pyqt.sourceforge.net/Docs/PyQt5/





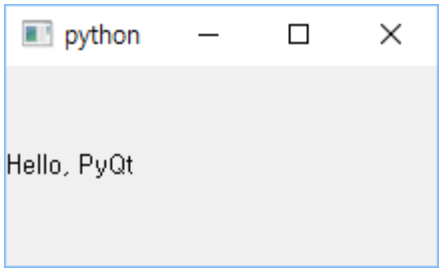
Hello, PyQt

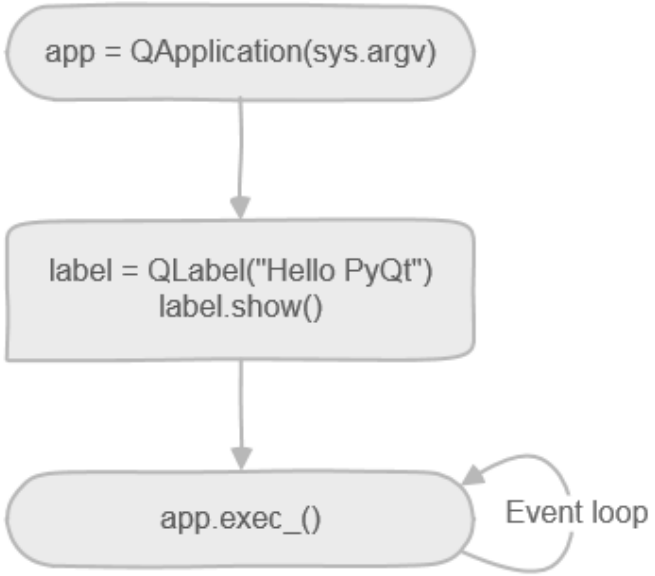
- In PyCharm
  - Create a new project "GUIwQt"
  - Create a new python file HelloQt.py

```
import sys
from PyQt5.QtWidgets import *

app = QApplication(sys.argv)
label = QLabel("Hello PyQt")
label.show()
app.exec_()
```

run

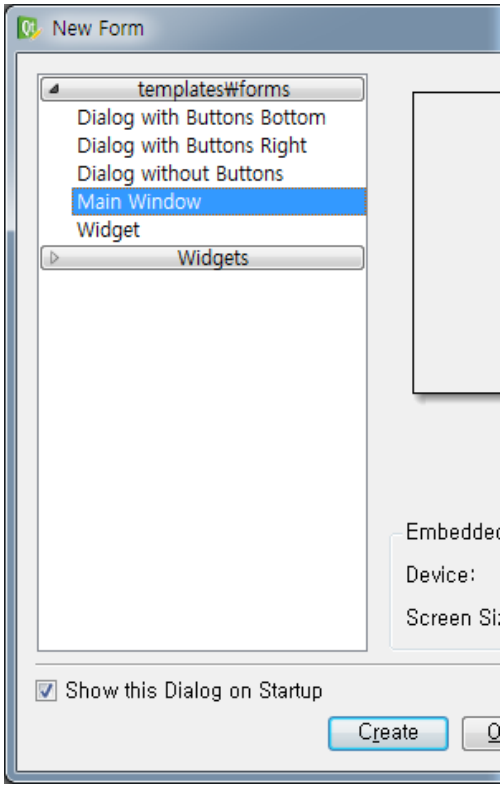


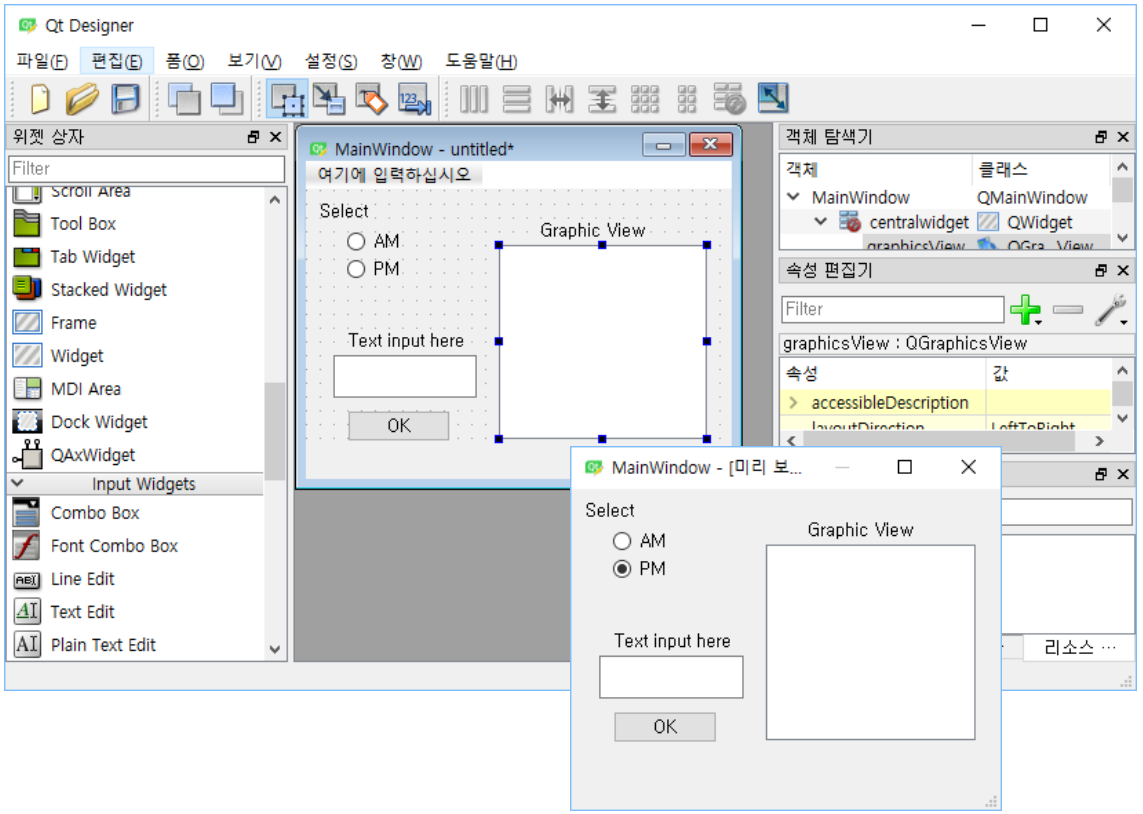


Qt Designer

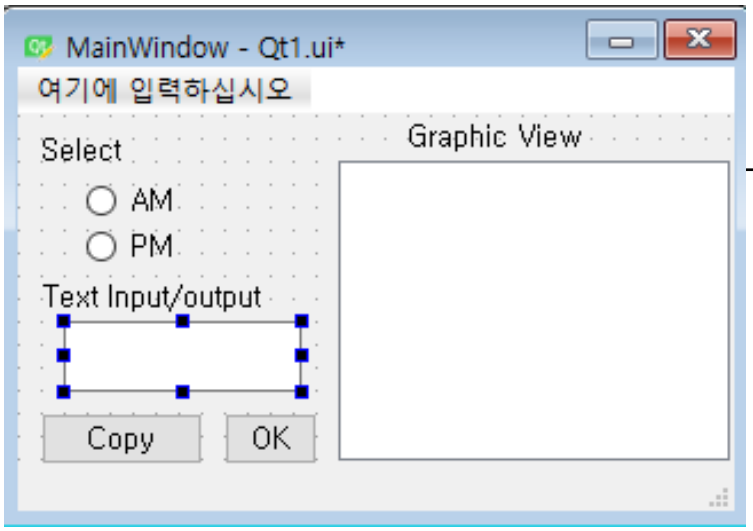
https://wikidocs.net/5225

- Qt Designer : Anaconda3/Library/bin/designer.exe
  - File > New Form > Main Window
  - Drag & Drop controls (widgets)
  - Test run : Control + R
  - File>Save → "Qt1.ui"

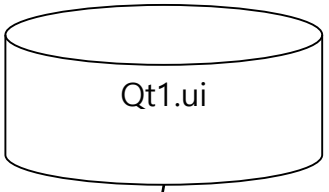




Bringing *.ui into Python



save




```
import sys
from PyQt5.QtWidgets import *
from PyQt5 import uic

class MyWindow(QMainWindow):
    def __init__(self):
        super().__init__()
        uic.loadUi('Qt1.ui', self)
        self.show()

if __name__ == '__main__':
    app = QApplication(sys.argv)
    window = MyWindow()
    sys.exit(app.exec_())
```

Connecting event handlers

```
import sys
from PyQt5.QtWidgets import *
from PyQt5 import uic

class MyWindow(QMainWindow):
    def __init__(self):
        super().__init__()
        uic.loadUi('Qt1.ui', self)
        self.copyButton.clicked.connect(self.copy_button_clicked)
        self.okButton.clicked.connect(self.ok_button_clicked)

    def copy_button_clicked(self):
        ampm = "Unselected"
        if self.amButton.isChecked():
            ampm = self.amButton.text()
        elif self.pmButton.isChecked():
            ampm = self.pmButton.text()
        self.textInputHere.setText (ampm)

    def ok_button_clicked(self):
        s = self.textInputHere.toPlainText()
        QMessageBox.about(self, "Message title", s)

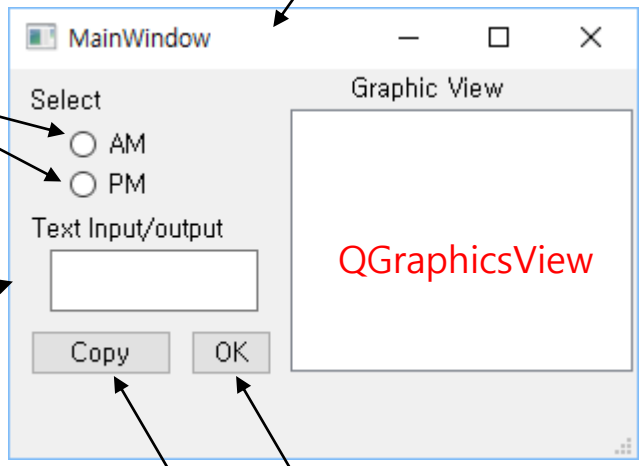
# main program
if __name__ == '__main__':
    app = QApplication(sys.argv)
    window = MyWindow()
    window.show()
    sys.exit(app.exec_())
```

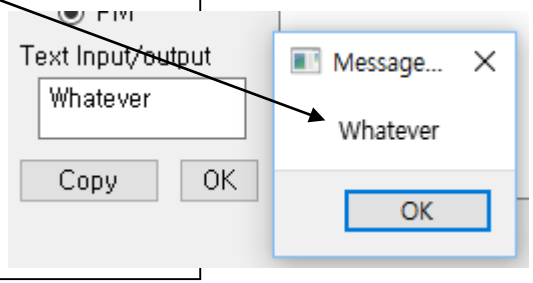
MyWindow (QMainWindow)

QRadioButton
: amButton, pmButton

QTextEdit
: textInputHere

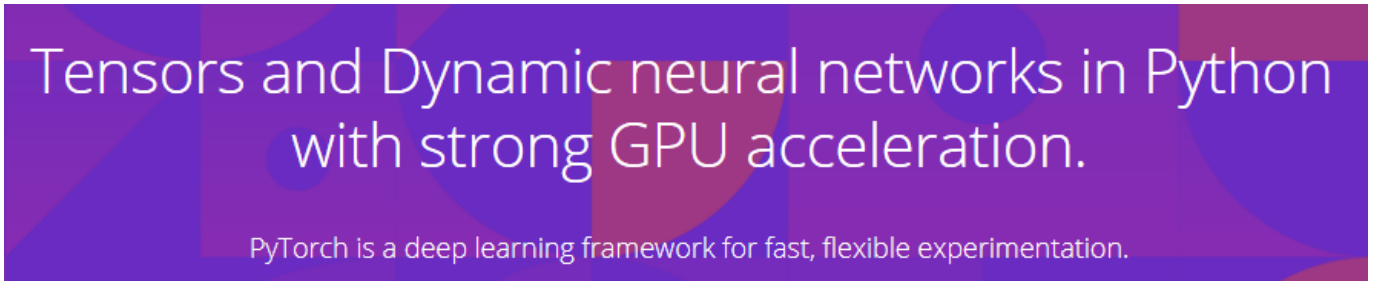
QPushButton
: copyButton, okButton





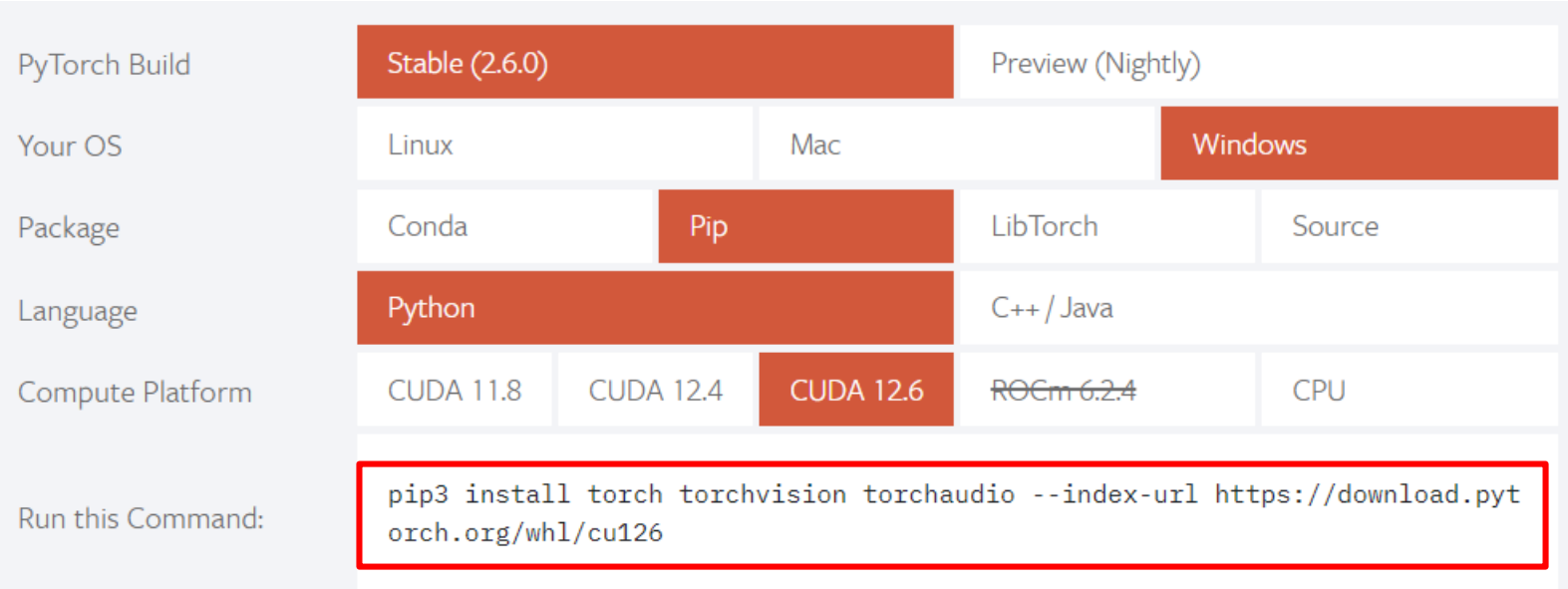
PyTorch

- Popular deep learning frameworks
  - TensorFlow (Google) ← Theano (U. Montreal)
  - PyTorch (Facebook) ← Torch (NYU / Facebook)
  - Caffe2 (Facebook) ← Caffe (UC Berkeley)
  - CNTK (Microsoft)
  - MXNet (Amazon)
  - …



Installing Pytorch on Windows and Anaconda

- Install required softwares
  - Pytorch : https://pytorch.org/
  - OS: Windows / Linux / Mac



Copy and execute this command